# SUSTAIN-A-THON (2.0) 2026

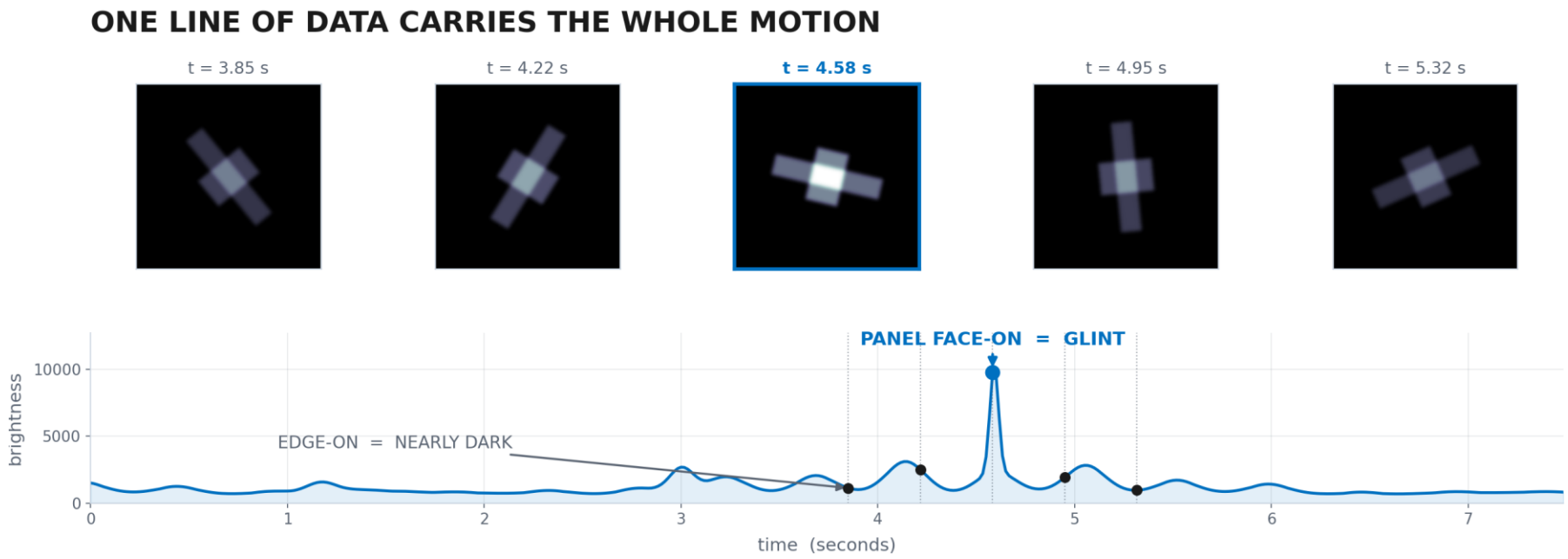
**SPINDEX — Spin Determination by INDEX of Reflected Light**

- **Problem Statement Title- Spin-state determination of uncooperative orbital debris**
- **Theme- Space Technology & Sustainability**
- **PS Category- Hardware**
- **Team ID- S080**
- **Team Name- ZEIT v1.0 Team Lead- Sarthak Tyagi**

# SPINDEX — SPIN FROM REFLECTED LIGHT

## ❖ Proposed Solution

- Reflected sunlight rises and falls as the target turns
- That light curve carries rotation, precession and cone angle
- Passive — no transponder, no cooperation, no contact
- Scored against a firmware-commanded rate, so it is falsifiable

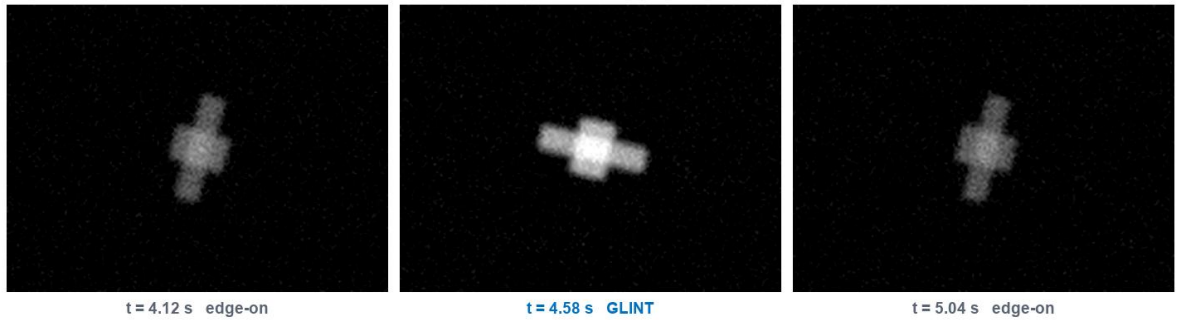



**Rendered target at five rotation phases — the flash at t = 4.58 s is a panel turning face-on**

# TECHNICAL APPROACH

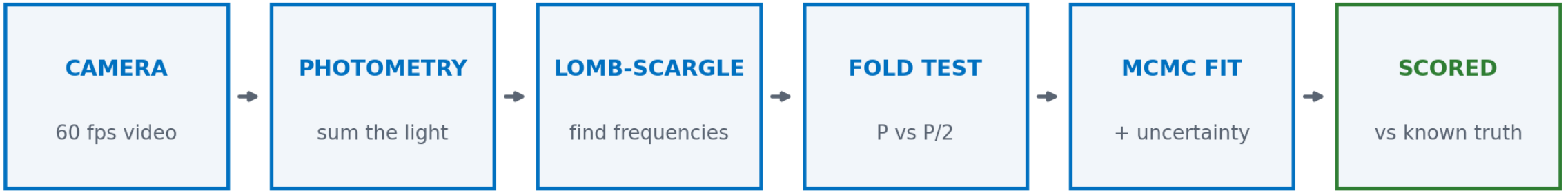
## ❖ Technologies and Methodology

- Software — Python, NumPy, SciPy, OpenCV, emcee
- Hardware — Arduino Uno, CNC Shield V3, 2 x NEMA-17, 10 W LED
- Photometry → Lomb-Scargle → fold test → MCMC fit
- Lomb-Scargle, not FFT: live capture is unevenly sampled




Frames from a recorded run

**60 fps in — spin rate, precession and cone angle out, with error bars**




Six-stage inversion pipeline, running live on arriving frames

# FEASIBILITY AND VIABILITY

## ❖ Feasibility, Risks and Mitigation

- Working prototype already built for ₹13,850
- 78 / 80 robustness runs and 44 / 44 unit tests pass
- Driver cogging → TMC2209 with 256-step interpolation
- Auto-exposure → locked, verified by a pixel-response test
- P vs P/2 ambiguity → sub-harmonic fold walk

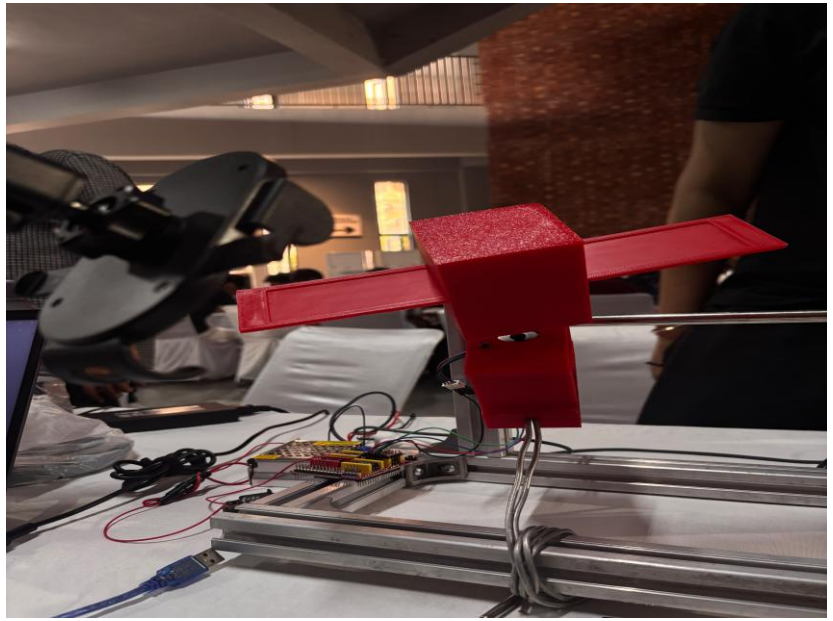
| PARAMETER | COMMANDED | RECOVERED | ERROR |
|---|---|---|---|
| **Rotation period** | 1.84971 s | **1.84971 s** | 0.000 % |
| **Precession period** | 11.40820 s | **11.40816 s** | 0.000 % |
| **Cone angle** | 60.000° | **59.976°** | 0.041 % |
| **Panel : bus area** | 0.09501 | **0.09524** | 0.237 % |

**Fitted to a light curve measured through real video compression — not to the model's own output**

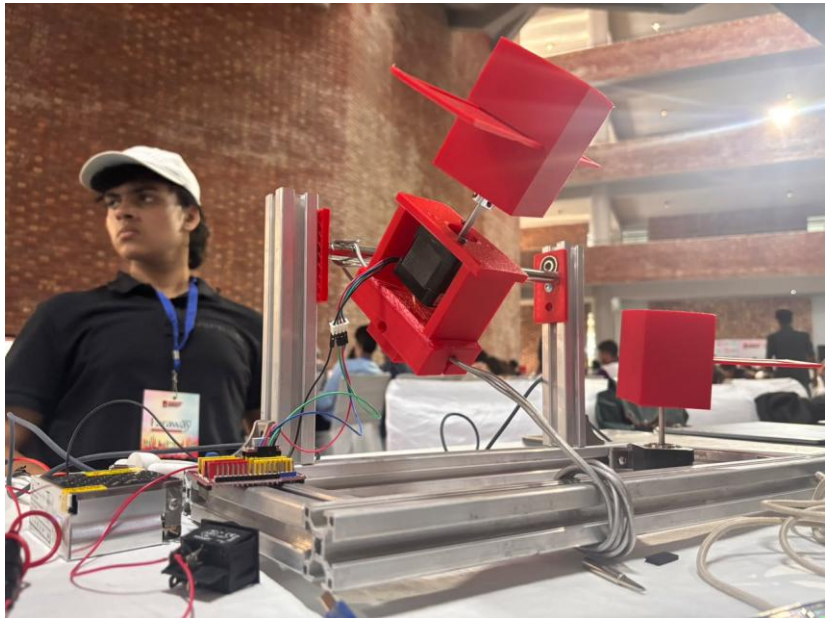
# IMPACT AND BENEFITS

## ❖ Impact and Benefits

- 36,000+ tracked objects; 5–10 must be removed each year
- Almost none of them has a known spin state today
- Environmental — prevents fragmentation during capture
- Economic — one camera screens a catalogue, not one object
- Users — servicers, SSA agencies, satellite operators



Box-wing target on the spin axis

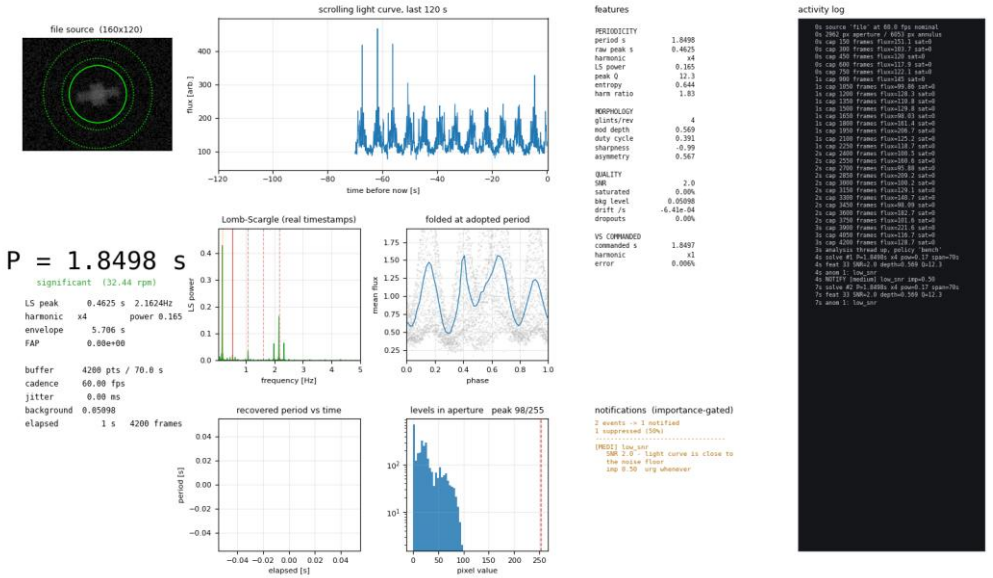


Cradle carrying the spin motor

# RESEARCH AND REFERENCES

## ❖ Research and References


- Zechmeister & Kürster (2009) — Generalised Lomb-Scargle. A&A 496, 577
- Stellingwerf (1978) — Phase dispersion minimisation. ApJ 224, 953
- Foreman-Mackey et al. (2013) — emcee. PASP 125, 306
- ESA Space Environment Report — debris population statistics
- Astroscale ADRAS-J (2024) — rendezvous with a spent upper stage
- Source and results — github.com/Sarthakk1407/SPINDEX


**Live monitor — light curve and running period estimate**